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Computer Engineering & Application

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Syntax of HTML Tags

Each HTML tag begins with a pair of opening angular bracket (<>). Between these brackets follows the name of the tag.

The end of the tag is denoted by a pair of closing angular bracket (</>) that has the name of the tag between.

Inside these set of brackets follow the content of the tag.

<HTML_TAG>

<!-- Tag's content -->

</HTML_TAG>

Example of HTML Tags

For instance, the <head> as explained later, forms the head of an HTML document and is used like this:

<head>

....

</head>

Content Inside an HTML Tag

The content between the closing and opening HTML tags constitute the behaviour of that particular tag.

For instance, an <h1> tag denotes a large heading. Content inside this tag will have the properties of the <h1> tag.

<h1>This is a large heading </h1>

Essential HTML Tags

There are some tags in HTML that are necessary for your HTML document because they present important information about our webpage for the interpretation of the browser.

Some examples of essential HTML tags include `<!DOCTYPE html>` tag, the `<head>` tag, the `<title>` tag, etc.

The `<!DOCTYPE html>` Tag

The `<!DOCTYPE html>` tag is the first tag of any HTML document. It's a special tag that does not have a closing set of brackets and doesn't have any HTML content inside it.

It is used only to tell the browser that the document loaded is an HTML document.

Here's the syntax to use it:

```
<!DOCTYPE html>
```

...

The `<html>` Tag

The first meaningful tag of any HTML document is the `<html>` tag. It indicates the scope of an HTML document.

If you put an HTML tag outside the `<html>` tag, it may not be considered by the browser.

Syntax

```
<html> ... </html>
```

The `<head>` Tag

The `<head>` tag appears at the top of any HTML document. It displays some other tags inside it that are used to give some additional information about the webpage.

For instance, **it may have `<meta>` tag to provide some meta information about the document as shown below.**

Syntax

```
<head>  
  <meta charset="UTF-8">  
</head>
```

The `<title>` Tag

The `<title>` tag is used to give a title to the HTML document. This title also appears on the tab of a browser where your HTML document is rendered.

Syntax

```
<title> HTML Tags </title>
```

The <body> Tag

The <body> tag encloses all HTML tags that are used to create your various HTML documents and webpages of your website.

Syntax

```
<body>  
...  
</body>
```

For instance, let's say you have a portfolio website where you want to tell about yourself. The entire content of your portfolio website would go inside the <body> tag as shown below:

```
<body>  
  
  <h1>  
  
    Hi there! This is my portfolio  
  
  </h1>  
  
  <p>I'm a <strong>web developer</strong></p>  
  
</body>
```

Thus the main content of your website is always present inside the <body> tag.

Other HTML Tags

Now that we're through with the basics, let's explore some other commonly used HTML tags alongside their usecases and syntax.

Unclosed HTML Tags

Some tags in HTML do not require a closing tag. They don't allow any content inside them.

Since these tags don't have a set of closing brackets, they're called as **Unclosed HTML tags**. They're also known as void elements in HTML.

For instance, the horizontal ruling tag <hr> is an unclosed HTML tag.

Syntax

```
<hr>
```

HTML Text Tags

Majority of content on any website comprises of texts. HTML provides a bunch of tags that allows us to render text on a webpage in different formats and sizes.

The <p> Tag

The <p> tag stands for the paragraph tag. It is used to render a paragraph or a group of words on the webpage.

```
<p>  
  This is a paragraph!  
</p>
```

You can use a <p> tag whenever you need to render more than one word, a line or a bunch of lines that need to be styled in a similar fashion.

The Tag

The tag is used for a group of words within a paragraph that need to be styled differently.

Here's an example of a tag that has a different font color than the rest of the paragraph:

```
<p>  
  This is going to be a paragraph.  
  
  Amidst the paragraph is <span style="color:gray"> this sentence styled differently.</span> Rest  
  of the paragraph is  
  
  the same.  
</p>
```

Now all the words enclosed within the tag will have a gray color, unlike those outside the tag as shown:

The Heading Tags

In a blogging website, you may need a title to display the blog title that has a large font-size. Or you may need to display titles of individual sections of the blog in a smaller font-size.

Thus, there are numerous such cases where you need to use headings. HTML has 6 different heading tags that you can use to render headings of different sizes.

```
<h1> This is H1 </h1>  
  
<h2> This is H2 </h2>
```

<h3> This is H3 </h3>

<h4> This is H4 </h4>

<h5> This is H5 </h5>

<h6> This is H6 </h6>

The <h1> is the largest heading, followed by <h2>, then <h3> till <h6> which is the smallest heading. Here's how the above code will look on your browser:

The Tag

Sometimes you need to lay more emphasis on some part of your paragraph. Browsers by default render anything inside a tag as bold text.

```
<p>  
  Hi there! My name is <strong>Siddhant</strong>  
</p>
```

In the above paragraph, the text inside the tag, ie, "Siddhant" will appear bold as shown:

The Tag

Just like the , the tag is also used to lay emphasis on certain text within a paragraph.

However, instead of making the text bold, browsers render content inside the tag as italics.

```
<p>  
  I am a web developer with more than  
  <em>  
    2 years of experience  
  </em>  
</p>
```

So the above would render the content inside tag as italic:

HTML Link Tags

There are two types of Link tags in HTML:

- The <a> or anchor tag to include hyperlinks to same pages of a website or different pages of another website.
- The <link> tag used for linking an internal or external resource to your HTML document.

Let's learn a bit about each of these now.

The <a> Tag

The <a> tag is used to add external or internal links as content in your HTML document.

Let's you want to display a menu or a navbar that links different pages of your website together. You can use a <a>tag to add these links.

```
<a href="/">Home</a>
```

```
<a href="/about">About</a>
```

```
<a href="/contact">Contact</a>
```

Above would render three internal links of your website as shown:

You could also include external links to other websites in your HTML document. Here's an example that demonstrates this:

```
<a href="www.google.com">Visit Google</a>
```

The <link> tag

HTML gives us the <link> tag which can be used to include an external asset, resource or file in a document.

For instance, you can use it to include a stylesheet in your HTML document like this:

```
<link rel="stylesheet" href="styles.css">
```

Above will link the styles.css file with your HTML document. Remember, <link> tag is an unclosed tag.

HTML Image and Object Tags

The tag in HTML is used to output or render images on the webpage. It specifies the source of the image using the src attribute as shown below:

```

```

The above would render the image specified inside the src attribute:

The <object> tag is used to render any external resource like an image, video, website inside a container in your HTML page.

You can specify the path of the resource inside the data attribute of the <object> tag as shown:

```
<object data="https://post.medicalnewstoday.com/wp-content/uploads/sites/3/2020/02/322868_1100-800x825.jpg" >
```

******The above would output the same image that the `<img>` tag did previously.

The `<object>` isn't used widely in modern websites. There are other alternatives to `<object>` tag like `<img>`, `<video>`, `<audio>` etc that are used in its place.

HTML List Tags

Lists are used quite often in websites to display a set of data or items in an ordered or unordered fashion.

HTML provides us with three types of list tags:

- Unordered List
- Ordered List
- Description List

Let's learn a bit about each of these.

Unordered List

An unordered list is represented with a `<ul>` tag. Inside a pair of `<ul>` tags, each list item is present using the `<li>` tag.

Each `<li>` tag or list item inside an unordered list is rendered with a bullet point icon.

```
<ul>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ul>
```

Ordered List

Similar to unordered list, ordered list or the `<ol>` tag is used to render a list of items that are marked with numbers instead of bullets:

```
<ol>
  <li>Item 1</li>
  <li>Item 2</li>
  <li>Item 3</li>
</ol>
```

Description List

HTML provides support for description list using the `<dl>` tag. A description list renders a list of terms using the `<dt>` tag.

For each term, we can use the `<dd>` tag to present the description of that term.

Consider the example below:

```
<dl>
  <dt>Term 1</dt>
  <dd>- Description of Term 1</dd>
  <dt>Term 2</dt>
  <dd>- Description of Term 2</dd>
</dl>
```

HTML Table Tags

HTML provides us with the `<table>` tag that renders some content in a table on the webpage.

We can create rows and columns to specify the structure of the table using the `<tr>` tag and the `<th>` tag. Then, we can display content inside each cell of the table using the `<td>` tag.

Consider the HTML snippet code below:

```
...
<style>
  table,
  th,
  td {
    border: 1px solid black;
    border-collapse: collapse;
  }
</style>
...
<table>
  <tr>
    <th>Name</th>
    <th>Role</th>
  </tr>
  <tr>
    <td>Siddhant</td>
    <td>Frontend Developer</td>
  </tr>
  <tr>
    <td>James</td>
    <td>Backend Developer</td>
  </tr>
</table>
```

HTML Form Tags

Forms are a common way to take multiple inputs from a user. For instance, a signup form on websites like Netflix takes a user's name, email and password. Similarly, a form that takes information to book a hotel for a user like the one you see on Oyo and Airbnb.

In such scenarios, HTML provides us with the `<form>` tag that we can use to create a form on the webpage. Inside the `<form>` tag, we can render input fields using the `<input>` tag and specify their labels using the `<label>` tag.

Here's an example of a form that takes the name and age of a user:

```
<form>
  <label for="name">Name:</label>
  <input type="text" id="name" name="name">
  <label for="age">Age:</label>
  <input type="text" id="age" name="age"><br><br>
</form>
```

HTML Scripting Tags

Modern websites are made highly interactive with JavaScript.

HTML allows us to use JavaScript on any HTML Document using the `<script>` tag as shown:

Syntax

```
<script>

</script>
```

Self Closing Tags

Self-closing tags are special tags in HTML that do not have any content, much like unclosed tags discussed earlier.

However, the difference is that self-closing tags have a backslash (/) in their syntax. HTML has two self-closing tags.

Here's an example of an `<svg>` tag that is self-closing in nature:

```
<svg height="100" width="100">

<circle cx="50" cy="50" r="40" stroke="black" stroke-width="3" fill="red" />
```

Comments in HTML

Comments in programs are subtitles for developers! Comments are messages for developers that don't have any meaning for the browser.

We can add a comment in a HTML page using the opening comment keyword `<!--` and the closing one `-->` as shown below:

Syntax

```
<!-- This is a comment! -->
```

Attributes in HTML

Attributes are properties of HTML tags that provide some additional information on these tags.

For instance, the src attribute on the tag tells us the URL of the image the tag is supposed to render.

```

```

Or the type attribute in the <input> tag specifies the type of the input field:

```
<input type="number" />
```

HTML <body> Tag

We all know that HTML is basically used to build websites that are available on the internet. HTML consists of series of elements like <head> tag <title> tag, etc that are used to structure the web page. One such element that is a home for all the other contents of the webpage such as headings, paragraphs, images, tables etc is HTML <body> Tag.

All the content inside the <body> tag is only visible on the webpage.

For example

```
<html>
<head>
  <title>Title of the WebPage</title>
</head>

<body>
  <p> Everything inside the body tag is visible on the web page.</p>
</body>

</html>
```

As shown in the example, the <body> tag is always enclosed within the <html> tag and there can be only one <body> tag in the whole webpage. It is usually present after the <head> tag. The content in the tag is written between the opening (<body>) and closing (</body>) tags.

Syntax

```
<body> Contents .... </body>
```

Tag Specific Attributes

<body> tag has some attributes that change the way the web pages looks. Let's learn about a few of them.

Attribute	Description
bbgcolor	Colour of the Background of your page
alink	Color of text for links
background	Image to be used as a background
link	Color of text for links that are not visited
vlink	Color of text for visited links
text	Color of the text on your webpage
margins	Used to add margins to the webpage

Window Event Attributes

There are some window event attributes that are script based and used within the <body> tag.

Few examples are given below for your reference.

Attribute	Description
ononline	Script triggered when the browser starts to work online
onoffline	Script triggered when the browser starts to work offline
onload	Script triggered when the page is completely loaded
onerror	Script triggered when an error occurs
onpageshow	Script triggered when a user visits a page
onresize	Script triggered when the browser window is resized

HTML <body> Tag Usage

Below is an example of basic structure of a webpage with <body> tag.

```
<html>
<head>
  <title>Title of the WebPage</title>
</head>

<body>
  <h1>This is a heading</h1>
  <p>This is a paragraph. Everything inside the body tag is visible on the web page.</p>
</body>

</html>
```

Explanation

- The <body> element contains all the contents of the webpage like the heading, paragraph, images, tables etc. Here <h1> element specifies a heading and <p> element specifies a paragraph.
- The <html> elements contains all the contents of the webpage.
- The <head> element contains additional information about the HTML page like the page title.
- The <title>element specifies a title for the webpage which is shown in the browser's window.

Below is another example <body> tag usage with **bgcolor** attribute.

```
<html>
<head>
  <title>Title of the WebPage</title>
</head>

<body bgcolor = 'yellow'>
  <h1>This is a heading</h1>
  <p>This is a paragraph. Everything inside the body tag is visible on the web page.</p>
</body>

</html>
```

Explanation

The HTML bgcolor attribute is used to set the background color of an HTML element.

Syntax

```
<body bgcolor = 'value'> Contents .... </body>
```

The Attribute value can be anyone of the following:

- **colorname:** The background color is set by using the color name. For example “yellow”
- **hexnumber:** The background color is set using the color hex code. For example “#0000ff”.
- **rgbnumber:** The background color is set by using the RGB code. For example: “RGB(0, 153, 0)”.

Tags	Description
Comments	This element is used to add comments in HTML. While a browser does not render HTML comments, they are visible to anyone viewing the page's source code
Doctype	A doctype or document type declaration specifies the markup language used by the current page within the web browser.
<a>	The <a> tag creates a hyperlink to a web page, a file, an email, or another location on the same page with its href attribute.
<abbr>	The <abbr> tag tag is typically used for abbreviations or acronyms of longer words, such as www, HTML, HTTP, etc.
<address>	In <address> tag, you can find the contact details of the site owner or the author of an article. The information may include addresses, cellphone numbers, emails, website connections, and many more.
<area>	The <area> tags identify clickable or active regions on an image-map associated with hyperlinks.
<article>	In HTML, the <article> element represents a piece of content that can theoretically be distributed to other websites or platforms without being an integral part of them. An entire blog post, news article, or similar content can be displayed with the <article> element.
<audio>	With <audio> element, audio content can be embedded in an HTML document without the need for an additional plug-in like Flash.
<aside>	The <aside> elements define content which is related to the main content of a webpage, but does not constitute it.
	The elements tag in HTML specifies bold text without giving it any additional significance.
<base>	All relative URLs inside a document are given a common target address by the <base> tag.
<bdi>	<bdi> tells the browser's bidirectional algorithm to isolate the text contained within it from other text in the document.
<bdo>	Overriding the current text direction is done with the <bdo> tag (short for bidirectional override).

Tags	Description
<blockquote>	<blockquote> tags provide a way to insert a quotation in the document from another source. Traditionally, browsers render the text inside a quote tag as indented text.
<body>	<body> tags in HTML are responsible for defining the main content within a page. This can include text, paragraphs, headings, images, tables, links, and videos.
 	A line break is produced by the tags. This is usually used in poems or addresses that require line breaks
<button>	Clickable buttons are represented by the <button> HTML element, and can be used in forms or anywhere in a document.
<canvas>	In documents,<canvas> elements define regions that may be used for drawing graphics through scripting (usually JavaScript).
<caption>	Table captions are created using the HTML <caption> tag. <table> tag is used with this element to represent a table's title.
<cite>	References to cited creative works are described by the <cite> HTML element, which must include the title of the cited work.
<code>	An HTML code fragment is represented by the <code> tag, which might be a snippet of HTML or CSS.
<col>	Column properties are specified by the <col> tag for each column within a <colgroup> element
<colgroup>	<colgroup>tag specifies the group of one or more columns. By doing so, you can apply styles to entire columns, rather than repeating a style for every column and every row
<data>	A machine-readable translation is added with the help of the <data> tag.Using this element, the processors can generate a machine-readable value and the browser can produce a human-readable value.
<datalist>	The<datalist> tags provide an autocomplete feature to form elements. Users can choose data from a list of predefined options.
<dd>	<dd> represents 'definition description' in HTML and is used to describe or define an item in list of descriptions
	Thetag indicates a deletion of a section of text from a document. Web browsers render the deleted text as strikethrough text.
<details>	A widget created with the <details> element shows information only when it is toggled into the "open" state.

Tags	Description
<dfn>	A definition element in HTML is represented by a <dfn> tag, and it's used to represent instances of definitions. The defining instance of a term is usually the first time the term appears in a document
<dialog>	The <dialog> element defines dialog boxes and other interactive components of a web page that can be interacted with by the user, such as dismissible alerts.
<div>	The <div> tags are generic containers that can hold flow content, which does not have a default rendering.
<dl>	The <dl> element of HTML represents a list of descriptions. A glossary or metadata can be displayed using this element
<dt>	A term in a description or definition list is indicated by the <dt> HTML element, which must be used within a <dl> element. A <dd> element normally follows it.
	An HTML tag emphasizes a particular word or phrase within a sentence. It renders in italic form on the browser and gives semantic meaning to the text within it.
<embed>	An HTML document can use the <embed> element to embed an external application, such as audio or video content.
<fieldset>	<fieldset> tag is used to create a group of related elements in a form, as well as to create a box around the elements
<figcaption>	In the <figure> element there is a <figcaption> element used to attach a caption to the image, table, or chart contained within the <figure> element.
<figure>	<figure> symbolizes an element that represents a unit of self-contained content, optionally with a caption, that is normally referred to as a single entity.
<footer>	The <footer> HTML element represents the footer of the sectioning root element or its nearest sectioning content element.
<form>	Using the <form> tag, HTML forms allow users to submit information to a website.
<head>	An <head> tag specifies the head section of a document, which contains document-specific information.
<header>	<header> tag is used to define the page's header, which contains information about the document or section's title and heading.
<hgroup>	<hgroup> tags in HTML represent heading groups and are used to group heading elements, such as headings and subheadings, from <h1> to <h6>.

Tags	Description
<h1> to <h6>	To define HTML headings, the heading tags are used. The browser displays headings large and bold by default. Aside from this, the <h1> headings are displayed in the largest font, whereas the <h6> headings are displayed in the smallest font.
<hr>	An HTML page can be divided or separated into sections with the <hr> tag, which stands for horizontal rule or theme break.
<html>	All HTML elements are contained within the <html> tag, which is the root of a HTML document
<i>	HTML uses the <i> tag to display italicized content. This tag is usually used to highlight an important term, phrase, or technical term.
<iframe>	With the iframe tag, another HTML document can be embedded within the current document.
	 tags are used to embed images in HTML pages. The HTML pages are linked to the images instead of directly inserting them in the document.
<input>	Forms with <input> tags enable interactive controls. Input fields can be of various types depending on the attribute type
<ins>	A block of inserted text is specified by the <ins> tag in HTML. It signifies that a range of text been added to the document
<kbd>	<kbd> tag refers to the textual part of inline text that represents the user's keyboard, voice, or any other text input device.
<label>	The <label> tag in HTML is used in order to provide improved usability for mouse users, i.e., clicks on the label will toggle the control.
<legend>	The title of the child contents is defined by the legend tag. Legends are parent elements. An element with this tag defines the caption for a <fieldset>.
	A list item is represented by the HTML element. This element must be contained in one of the following parent elements: an ordered list (, an unordered list , or a menu)
<link>	An external resource is linked to the current document using the <link> tag.
<main>	Typically, the <main> element specifies the most important or dominant content within the <body> of a document.
<map>	An image map is defined by the <map> tag. Image maps have clickable areas.
<mark>	A <mark> tag represents text that is marked or highlighted for reference or notation purposes, based on its relevance and importance in the context of the enclosing element.

Tags	Description
<meta>	The HTML document's metadata is represented by the <meta> tag. Among the information it contains are the page description, keywords, copyright, the language used, and the author.
<meter>	Measurements within a given range or fractional values are represented by the <meter> element. Alternatively, called a gauge.
<nav>	A <nav> element represents a section of navigational links in a document.. Menus, table of contents, and indexes are a few examples of navigation links.
<noscript>	Using the <noscript> tag, you can provide alternate content for users who don't support scripts in their browsers, or have a browser that doesn't support client-side scripting, or have a browser that doesn't support scripts in any form.
<object>	The <object> tag is an HTML tag and used to display multimedia like audios, videos, images, PDFs, and Flash in web pages.
	The tag is for an ordered list, an ordered list can be numerical or alphabetical. Inside the tag you have to make a list of items that will follow the order.
<optgroup>	Within a dropdown list defined by the <select> element, the <optgroup> tag defines a group of options.
<option>	Using the <option> element, you can define an option on a dropdown list defined by the <select> element.
<output>	The<output>HTML element is a container element into which a site or app can inject the results of a calculation or the outcome of a user action.
<p>	Paragraph is defined by the<p> tag.Paragraphs are usually represented in visual media as blocks of text separated from adjacent blocks by blank lines and/or first-line indentation, but HTML paragraphs can be any structural grouping of related content, such as images or form fields.
<param>	Parameters are defined by the <param> tag for an <object> element. It is used for passing parameters to an embedded object
<picture>	The <picture> tag in HTML is used to give flexibility to the web-developers to specify image resources. The <picture> tag contains <source> and tags
<pre>	<pre> tag in HTML are used to describe the block of preformatted text, which preserves formatting characters such as spaces, line breaks, tabs, etc.
<progress>	A progress element indicates how far a task has progressed.Using it, web developers can easily create a progress bar on their websites.

Tags	Description
<q>	An inline quotation is marked by the <q> HTML element. The text is surrounded by quotation marks in most browsers today.
<rp>	When a browser does not support ruby annotations, it will use the <rp> element as a fallback parenthesis.
<rt>	The <rt> tag defines a way to explain or pronounce characters in a ruby annotation.
<ruby>	<ruby> is usually used to indicate East Asian characters' pronunciations, and represents small annotations above, below, or next to the base text.
<s>	Using the <s> HTML element, you can render text with a strikethrough or a line through it. Things that are no longer accurate or relevant should be represented in the <s> element
<samp>	<samp> may be used to specify the output sample from a computer program. It's Content is typically displayed using a monospaced font in the browser
<script>	The script tag allows for the insertion of a client-side script (JavaScript).It enables you to insert a script within your HTML document.
<section>	Whenever two headers or footers are required or any other section of the document is needed, the <section> tag is used.
<select>	A list of options is defined by nesting the <option> element within the <select>element.
<small>	If you use the <small> tag, you can reduce the font size by one size smaller than its base size; for instance, from large to small.
<source>	For media elements like <audio> and <video>, the <source> element specifies multiple alternative media resources.
	Elements with attributes indicate that their contents are highly important, serious, or urgent. Typically, browsers display the content in bold.
	 is a generic inline HTML element for phrasing content, which inherently represents nothing. Styles can be applied to it by grouping elements together.
<style>	Style information (CSS) can be defined for a document using the <style> tag.
<sub>	<sub> elements are used to specify inline text that should be displayed as subscript purely for typographical reasons.
<summary>	A summary, caption, or legend for the disclosure box of a <details> element is specified by the <summary> HTML element. By clicking the <summary> element, you can toggle the open/closed

Tags	Description
	state of the parent <code>*details></code> element.
<code><sup></code>	Inline text can be displayed as a superscript with the <code><sup></code> HTML element only for typographical reasons
<code><svg></code>	HTML documents can embed SVG graphics using the <code><svg></code> element. With this tag you can define vector graphics in an XML format.
<code><table></code>	Tabular data is represented by the <code><table></code> HTML element, i.e., information contained in a two-dimensional table comprised of rows and columns of cells.
<code><tbody></code>	The <code><tbody></code> element encapsulates a set of table rows (<code>*tr></code> elements), which are associated with the table body (<code><table></code>).
<code><td></code>	<code><td></code> tags define a table cell. A table row may consist of more than one <code><td></code> element. The default alignment is left in this element.
<code><template></code>	<code><template></code> elements are used to define HTML fragments that are not in immediate need of rendering, but can simply be cloned and inserted into documents by JavaScript.
<code><textarea></code>	Using the <code><textarea></code> tag, a multiline text input control is created within a form.
<code><th></code>	Using the <code><th></code> HTML element, a cell is defined as the header of a group of cells in a table. The scope and headers attributes determine the exact nature of this group.
<code><thead></code>	A table's header is defined by the <code><thead></code> element .This element is a child of the <code><table></code> element along with the <code><thead></code> and <code><tfoot></code> tags
<code><time></code>	Displaying a human-readable date/time is done by using the <code><time></code> tag. In addition, date and time can also be encoded.
<code><tr></code>	A row can then be constructed by combining <code><td></code> (data cells) and <code><th></code> (header cells).
<code><track></code>	Text tracks such as subtitles and captions can be specified as <code><track></code> elements for <code><audio></code> and <code><video></code> elements.
<code><u></code>	<code><u></code> elements represent inline text that contains non-textual annotations and should be rendered accordingly.
<code></code>	An <code></code> tag is suitable for elements that do not need to be presented numerically and which can be rearranged without changing the meaning of the list.
<code><var></code>	A <code><var></code> tag indicates that the text is an example of a variable or program argument.

Tags	Description
<video>	A <video> tag is intended for embedding video content into a document, such as a movie clip or a video stream.
<wbr>	<wbr> tags indicate in which part of a text it would be okay to add a line break.

The tag is widely used to output images on the web page. It can be styled flexibly like any other HTML element. It can also be used with the <picture> tag, which is used to output multiple images based on certain conditions.

Scope of Article

- The article talks about the tag, its uses and commonly used attributes.
- The article also explains the <picture> tag, how and why it is used.
- The article presents the browser support offered for the <picture> tag.

< img > Tag

HTML allows us to easily render and output images on a web page using the tag. It creates a container on the web page to hold a reference to the image that you specify.

Example

To render an image on a page using the image tag, you have to add the URL of the image inside the src attribute of the img tag. Here's how you can do that:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Image Tag in HTML!</title>
</head>
<body>
  
</body>
</html>
```

Usage

The tag is used wherever you want to display an image on your web page. You can also combine it with other HTML elements to create more complex UI components. For instance, you can attach an image inside a card to make a profile card.

Attributes of the Tag

Some of the common attributes of tag include:

- **src**: Provides the path or the URL to the image being rendered.
- **height**: Sets the height of the rendered image.
- **width**: Sets the width of the rendered image.
- **alt**: Provides an alternate text to the image that is displayed if the image fails to load on the webpage.
- **loading**: Specifies how images should load when the webpage is rendered.

Examples

Let's look at how we can use the above attributes using some examples.

Use of Height and Width Attribute with Tag

The tag allows you to directly specify the dimensions of the rendered image using the height and width attributes.

```

```

The specified height and width are taken as a string and represent the height and width of the image in pixels. So in the above case, the image will be 100 pixels wide and have a height of 100 pixels as well.

Use of alt Attribute

There are a number of reasons a user may not be able to see an image that the tag is rendering. The image link could be broken; the user could be on a slow internet connection, etc.

In such cases, the alt tag provides a short description of what the image represents. It stands for the alternate text and is also considered useful for boosting the SEO scores of the images on your website.

```

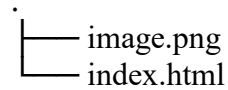
```

In the above example, if your image fails to appear, "JavaScript Logo" text will be rendered in its place to give the users an idea of what the image represents.

How to Get Image from Another Directory/Folder with Relative Path?

Besides using a URL, we can also use the relative path of an image. Consider the project directory shown in the form of the tree below and the explanation along with it to understand the relative path concept.

1. The tree shown below represents that image.png and index.html are in the same level(same directory of project).



0 directories, two files

Let's see how we will include the image.png in our index.html relatively.

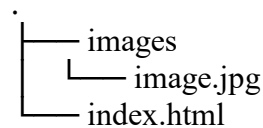
```

```

Explanation:

The ./ conveys that the file or directory is written after it exists in the same directory. So, we can access it by writing a single dot followed by a forwarding slash.

2. The tree shown below represents that the index.html file and images directory are at the same level and image.png is inside the images directory.



One directory, two files

Let's see how we will include the image.png in our index.html file,

```

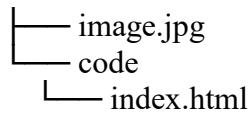
```

Explanation:

The ./ written in the path conveys that the images directory is located at the same level as index.html(which contains the img tag), and inside this directory, there exists an image.png which image tag should use to represent the corresponding graphics on the web.

3. The tree shown below represents that image.png exists on the root level of our project directory, and there also exists another directory named code which consists of index.html.

```
.
```



1 directory, 2 files

Let's see how we will include the image.png in our index.html file,

```

```

Explanation:

The ../ (double dot) in the path conveys that the file image.png is located on the one level back in the project structure.

Use < img > Tag as a Link

By wrapping it inside the anchor tag, you can also use the image tag as an internal or external link.

```
<a href="#">
  
</a>
```

When you click on the image, you will be redirected to the url specified in the href property of the anchor tag.

Use of Loading Attribute

You can specify the loading mechanism of the image using the loading attribute of the image tag.

If you set the loading attribute to lazy, the image will load only when it comes in the view of the webpage.

```
<img
  loading="lazy"
  ...
/>
```

If you want the image to be loaded instantly, you can use the eager value in the loading attribute instead.

```
<img
  loading="eager"
  ...
/>
```

HTML < picture > Tag

The <picture> tag is used to output or renders multiple images using the source tag or the tag.

It is commonly used in scenarios where you want to output different-sized images depending on the screen width or the dimensions of the size.

Example

Here's how you can use the <picture> tag to output multiple images:

```
<picture>
  <source media="(min-width:769px)" srcset="./image-large.png" />
  <source media="(min-width:465px)" srcset="./image-medium.png" />
  
</picture>
```

So in the above case, we will render image-large for all devices that are more than 769 pixels wide. For devices that are between 465 and 769 pixels wide, we'll render image-medium. Finally, we'll render images small for devices smaller than 465 pixels.

In case the browser does not support <picture/> tag or the url of <source> tag could not be resolved, the tag acts as a fallback. So the URL or the src of the is used to populate the area where the <picture/> was supposed to render an image on the webpage.

The tag is widely used to output images on the web page. It can be styled flexibly like any other HTML element. It can also be used with the <picture> tag, which is used to output multiple images based on certain conditions.

Scope of Article

- The article talks about the tag, its uses and commonly used attributes.
- The article also explains the <picture> tag, how and why it is used.
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Introduction to HTML < img > Tag

HTML allows us to easily render and output images on a web page using the tag. It creates a container on the web page to hold a reference to the image that you specify.

Example

To render an image on a page using the image tag, you have to add the URL of the image inside the src attribute of the img tag. Here's how you can do that:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <title>Image Tag in HTML!</title>
</head>
<body>
  <img
    <!-- Here, the myImage.png exists in the same directory. That's why the URL
    "myImage.png" is able to access that particular image -->
    src="myImage.png"
  />
</body>
</html>
```

HTML < picture > Tag

The <picture> tag is used to output or renders multiple images using the source tag or the tag.

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In case the browser does not support <picture/> tag or the url of <source> tag could not be resolved, the tag acts as a fallback. So the URL or the src of the is used to populate the area where the <picture/> was supposed to render an image on the webpage.

The <marquee> tag

The marquee tag in HTML is a container tag that is used for implementing scrollable text or images in a web document. It can scroll texts or images either from top to bottom (or vice-versa) or left to right (or vice-versa).

Note: This tag has been **deprecated** in HTML5

Syntax

```
<marquee> ... </marquee>
```

Attributes

Apart from **global attributes**, the marquee tag in HTML accepts the following attributes.

Attribute	Description
behavior	<i>defines the type of scrolling in the marquee tag in HTML, takes the values scroll, slide and alternate.</i>
direction	<i>defines the scrolling direction in which the marquee tag in HTML, takes the values up, down, left, right.</i>
height	<i>defines the height or length of a marquee, takes values in pixels or percent(%).</i>
loop	<i>defines how many times the marquee will loop. By default its value is infinite, i.e. the marquee will loop infinitely, takes values in the form of numbers.</i>
scrolldelay	<i>defines how long to delay between each jump, takes values in seconds.</i>
scrollamount	<i>defines how far to jump, takes values in the form of numbers.</i>
width	<i>defines the width or breadth of a marquee, takes values in the form of numbers.</i>
vspace	<i>defines vertical space around the marquee, takes values in pixels.</i>
hspace	<i>defines horizontal space around the marquee, takes values in pixels.</i>

Example

```
<!DOCTYPE html>
<html>

  <head>
    <title>HTML marquee Tag</title>
```

```
</head>

<body>
  <marquee> This text will go from right to left</marquee>
  <marquee direction = "down">This text will be go from top to bottom</marquee>
</body>

</html>
```

The <pre> tag

The pre tag in HTML is used to define preformatted text. The text item enclosed within the pre tag in HTML is formatted in a fixed-width font and the spaces and the line-breaks are preserved in the text, i.e. the text enclosed within the<pre> tag will be shown **exactly** as written inside the pre tag in HTML.

Syntax

```
<pre> ... </pre>
```

Attributes

The pre tag in HTML only accepts *global attributes*.

Note: Global attributes are attributes common to all HTML elements; they can be used on all elements

Example

```
<!DOCTYPE html>
<html>
<body>

<pre>
This is an example of      pre
tag in html, as
    we can see
    that
    this
document is not structured at all.
</pre>

</body>
</html>
```

The <i> tag

The i tag in HTML is used to display the text in italic style. The text placed within i tag in the HTML document gets *emphasised* by HTML.

Note: In the newer versions of HTML, the tag is generally preferred over i tag in the HTML document.

Syntax

```
<i> ... </i>
```

Attributes

It only accepts **global attributes**.

Example

```
<!DOCTYPE html>
<html>
<body>

<p> The i tag in HTML is used to write text in <i>italic style</i>. </p>

</body>
</html>
```

Explanation of the example:

In the above example, the text placed within the i tag in HTML gets emphasised in the web document.

The <figure> tag

The figure tag in HTML is used to store a photo or a group of photos, diagrams, code listing etc. It is used to store items that have **embedded content**. For e.g. we can store an <image> and <figcaption> within a figure tag in HTML, and both of them would be considered as one entity in the web document.

Note: The content within the figure tag in the HTML document is related to the main flow, but its position is independent of the main flow and does not affect the flow of the document when removed.

Syntax

<figure> ... </figure>

Attributes

The figure tag in HTML only accepts **global attributes**.

Example

```
<!DOCTYPE html>
<html>
<body>

<figure>

<figcaption> Official logo of HTML5 </figcaption>
</figure>
</body>
</html>
```

Explanation of the example:

In the above example, the **image** of the logo and the **caption** to the image is stored as one entity within the <figure> tag in the HTML document.

The <progress> tag

The progress tag in HTML defines the completion of the progress of the tag. It is displayed as a **progress bar** on the webpage.

Syntax

<progress />

Note: It is a self-closing tag.

Attributes

Apart from **global attributes**, the progress tag in HTML takes the following attributes:

Attribute	Description
max	<i>defines the upper limit of the progress bar. It is a floating point number that takes positive values.</i>
value	<i>defines how much a task has been completed. It is a floating point number that takes values between 0 and max-value</i>

Example

```
<!DOCTYPE html>
<html>
  <body>

    <h1>Year progress bar for april: </h1>
    <progress value = "4" max = "12"/>

  </body>
</html>
```